Topology, Edge States and Disorder in the Su–Schrieffer–Heeger Chain

Bulk invariants, bulk–boundary correspondence and the topological Anderson insulator

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Abstract

We present a numerical study of the Su–Schrieffer–Heeger (SSH) chain, the minimal one-dimensional topological insulator. Starting from the Bloch Hamiltonian we derive the winding number, the Berry connection and Zak phase, the zero-energy edge state and its localisation length $\xi = 1/\ln|w/v|$, and the finite-size splitting of the two end modes. All bulk quantities are computed by gauge-invariant discretisations that never differentiate an eigenvector: the winding number is an integer to 2.5×10^{-16} and the Zak phase equals $\pi\nu$ to 1.3×10^{-15}, unchanged under a random gauge. The numerical edge state reproduces ξ to round-off. We derive the edge splitting $\varepsilon(N) \simeq w(1 - v^2/w^2)(v/w)^N$ from the overlap of the two end states, which corrects the commonly quoted prefactor $1-(v/w)^2$ by a factor w. With chiral-symmetric hopping disorder we evaluate the real-space winding number, derive the transition from the zero-energy transfer matrix, $\langle \ln|v_n| \rangle = \langle \ln|w_n| \rangle$ ($W_c = 3.959$), observe a disorder-induced topological Anderson insulator, and use finite-size scaling to show that equal disorder on both hoppings never destroys the topological phase.

1 Introduction

Topological insulators are band insulators whose ground state carries a quantised invariant that cannot change without closing the gap. The physical consequence, *bulk–boundary correspondence*, is that an interface between phases with different invariants must host gapless states. The SSH chain [1], originally a model of polyacetylene, is the simplest system in which all of this can be worked out exactly: two sites per cell, two hoppings, and a chiral symmetry that quantises a winding number.

Because it is exactly solvable, the SSH chain is the natural testbed for numerical methods that are later applied to models with no closed form: gauge-invariant Berry phases, edge-state identification on finite chains, and real-space invariants for systems without translation symmetry. The aim of this project is not only to reproduce textbook results [2] but to check each numerical tool against an analytic answer to machine precision, so that the same tools can be trusted when disorder removes the analytic answer.

Contributions.

1. A gauge-invariant evaluation of the winding number and Zak phase (discrete phase sums and Wilson loops), checked against exact values and against random gauge transformations.
2. Identification of edge states on finite chains by diagonalising the chiral operator in the near-zero subspace, which gives sublattice-pure end states and the localisation length to round-off.
3. A derivation of the edge splitting from the overlap of the end states, giving the prefactor $w(1 - v^2/w^2)$ rather than $1 - (v/w)^2$.
4. A closed-form condition for the disorder-driven transition, compared with the disorder-averaged real-space winding number and finite-size scaling, including the topological Anderson insulator.

Table 1: Summary of the main results.

Quantity	Computed	Analytic
Winding number, 50 values of v/w	integer to 2.5×10^{-16}	0 or 1
Open chain $N = 60$, $v/w = 0.5$: $\min \lvert E \rvert$	2.2×10^{-16}	0
Open chain $N = 60$, $v/w = 1.5$: $\min \lvert E \rvert$	$0.504\, w$	$\approx \lvert v - w \rvert = 0.5\, w$
ξ at $v/w = 0.3, 0.5, 0.7$	0.830584, 1.442695, 2.803673	$1/\ln(w/v)$
Splitting slope, $v/w = 0.5$	-0.69316	$\ln(v/w) = -0.69315$
Splitting prefactor, $(v, w) = (1, 2)$	1.500	$w(1 - v^2/w^2) = 1.5$
Zak phase	π / 0, error $\leq 1.3 \times 10^{-15}$	$\pi\nu$
Zak phase under random gauge	unchanged to 2.5×10^{-15}	gauge invariant
Disorder transition, $v = 0.5$, $W_w = W_v/2$	ν drops between $W = 3.25$ and 4.5	$W_c = 3.959$
Topological Anderson insulator, $v = 1.2$	$\nu = 0.68 \pm 0.04$ at $W = 2.75$	$\nu = 1$ for $2.299 < W < 3.515$

2 Theory and derivations

2.1 Model

The chain has two sublattices, A and B, an intracell hopping v and an intercell hopping w, both taken real and positive unless stated otherwise:

$$H = \sum_n \Big[v\, c^\dagger_{A,n} c_{B,n} + w\, c^\dagger_{B,n} c_{A,n+1} + \text{h.c.} \Big]. \tag{1}$$

Ordering the basis as $(A_1, B_1, A_2, B_2, \dots)$, the single-particle Hamiltonian of an open chain of N cells is the $2N \times 2N$ tridiagonal matrix with alternating off-diagonal entries $v, w, v, w, \dots$ and zero diagonal.

2.2 Bloch Hamiltonian and spectrum

For a periodic chain we Fourier transform over cells,

$$c_{\alpha,n} = \frac{1}{\sqrt{N}} \sum_k e^{ikn}\, c_{\alpha,k}, \qquad \alpha \in \{A, B\}. \tag{2}$$

The intracell term is diagonal in k with coefficient v. The intercell term $c^\dagger_{B,n} c_{A,n+1}$ picks up e^{ik} from the shift $n \to n + 1$. Collecting terms,

$$H = \sum_k \begin{pmatrix} c^\dagger_{A,k} & c^\dagger_{B,k} \end{pmatrix} H(k) \begin{pmatrix} c_{A,k} \\ c_{B,k} \end{pmatrix}, \tag{3}$$

$$H(k) = \begin{pmatrix} 0 & h^*(k) \\ h(k) & 0 \end{pmatrix}, \qquad h(k) = v + w\, e^{ik}. \tag{4}$$

No intracell position phase is included, so $H(k + 2\pi) = H(k)$. The eigenvalues follow from $E^2 = |h(k)|^2$:

$$E_\pm(k) = \pm |h(k)| = \pm \sqrt{v^2 + w^2 + 2vw \cos k}. \tag{5}$$

The minimum of $|h|$ is at $k = \pi$, where $|h(\pi)| = |v - w|$. The gap is therefore $2|v - w|$ and closes only at $v = w$.

The d-vector picture. Any 2×2 traceless Hamiltonian can be written $H(k) = \mathbf{d}(k) \cdot \boldsymbol{\sigma}$. Here

$$d_x = \operatorname{Re} h = v + w \cos k, \qquad d_y = \operatorname{Im} h = w \sin k, \qquad d_z = 0. \tag{6}$$

As k runs over the Brillouin zone, $\mathbf{d}$ traces a circle of radius w centred at $(v, 0)$ in the $d_x d_y$ plane. The energy is $\pm|\mathbf{d}|$, so the gap closes exactly when the circle passes through the origin.

2.3 Chiral symmetry

Let $\Gamma = \sigma_z$ in each cell, i.e. $+1$ on A and -1 on B. Because H only connects A to B,

$$\Gamma H \Gamma = -H. \tag{7}$$

Two consequences follow.

1. If $H\psi = E\psi$ then $H(\Gamma\psi) = -E(\Gamma\psi)$, so the spectrum is symmetric about zero.
2. In the $E = 0$ eigenspace Γ commutes with H, so zero modes can be chosen as eigenstates of Γ, i.e. supported on one sublattice only.

In the Bloch picture chiral symmetry forces $d_z = 0$: a σ_z term would commute with Γ rather than anticommute. This confines $\mathbf{d}(k)$ to a plane, and closed loops in a punctured plane are classified by an integer. Together with time reversal and particle–hole symmetry (both present for real hoppings), the SSH chain is in class BDI; the winding number below is its $\mathbb{Z}$ invariant.

2.4 Winding number

The winding number counts how many times $h(k)$ encircles the origin:

$$\nu = \frac{1}{2\pi i} \oint_{\mathrm{BZ}} \frac{\partial_k h(k)}{h(k)}\, dk = \frac{1}{2\pi} \oint_{\mathrm{BZ}} \partial_k \arg h(k)\, dk. \tag{8}$$

Writing $z = e^{ik}$ turns the integral into a contour integral over the unit circle,

$$\nu = \frac{1}{2\pi i} \oint_{|z|=1} \frac{w}{v + wz}\, dz. \tag{9}$$

The integrand has a single pole at $z = -v/w$, which lies inside the unit circle iff $|v| < |w|$. By the residue theorem,

$$\nu = \begin{cases} 1, & |v| < |w|, \\ 0, & |v| > |w|. \end{cases} \tag{10}$$

At $|v| = |w|$ the pole sits on the contour, h vanishes at $k = \pi$, and ν is undefined. Since ν is an integer that depends continuously on (v, w) wherever it is defined, it can only change where the gap closes.

2.5 Berry connection and Zak phase

For $h \neq 0$ write $h(k) = |h| e^{i\phi(k)}$. The eigenvectors of Eq. (4) are

$$u_{\pm}(k) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm e^{i\phi(k)} \end{pmatrix}. \tag{11}$$

The Berry connection of the lower band is

$$A(k) = i\langle u_- | \partial_k u_- \rangle = \frac{i}{2}\big(0 + e^{-i\phi}\, \partial_k e^{i\phi}\big) = -\frac{\phi'(k)}{2}. \tag{12}$$

The Zak phase is its integral over the Brillouin zone:

$$\gamma = \oint A\, dk = -\frac{1}{2} \oint \phi'\, dk = -\pi\nu. \tag{13}$$

Under a gauge change $u \to e^{i\chi(k)} u$, $A \to A - \chi'$ and γ changes by $-[\chi(2\pi) - \chi(0)]$. Single-valuedness requires this to be a multiple of 2π, so γ is defined modulo 2π and

$$\gamma \equiv \pi\nu \pmod{2\pi}. \tag{14}$$

Zak phase as polarisation. By the King-Smith–Vanderbilt relation, the electronic polarisation of a filled band is $P = e\gamma/2\pi$ per cell [3]. $\gamma = \pi$ means the Wannier centre of the filled band sits half-way between unit-cell origins, i.e. on the intercell bond w. This is the bulk origin of the half-charges bound to the ends of the chain.

Wannier centre. The same result follows from the Wannier function of the filled band, $|w_R\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-ikR}|\psi_{-,k}\rangle\, dk$, where $|\psi_{-,k}\rangle = e^{ikX}|u_-(k)\rangle$. Its position expectation value is

$$\langle w_0|X|w_0\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} i\langle u_-|\partial_k u_-\rangle\, dk = \frac{\gamma}{2\pi} \pmod 1. \tag{15}$$

For $\nu = 1$ the centre sits at $\frac{1}{2}$ (mod 1), between the unit cells, while for $\nu = 0$ it sits at the cell origin.

Dependence on the origin. If the B site is placed at position x_B inside the cell and the Fourier transform includes that phase, $H(k + 2\pi) \neq H(k)$; the two conventions differ by $U(k) = \mathrm{diag}(1, e^{ikx_B})$. The Zak phase then shifts by πx_B, so for $x_B = \frac{1}{2}$ the two phases have $\gamma = \pm\pi/2$. Only differences of γ between phases, at fixed origin, are physical.

2.6 Zero-energy edge state

Consider a semi-infinite chain that starts at cell 0 with an A site, and look for a solution with $E = 0$ and $\psi_B(n) = 0$ for all n. The rows of $H\psi = 0$ are of two kinds.
B rows (the equation on site B_n):

$$v\,\psi_A(n) + w\,\psi_A(n+1) = 0. \tag{16}$$

This is a first-order recursion with solution

$$\psi_A(n) = \Big(-\frac{v}{w}\Big)^n \psi_A(0). \tag{17}$$

A rows (site A_n): $v\,\psi_B(n) + w\,\psi_B(n-1) = 0$, which is satisfied identically because $\psi_B \equiv 0$. The first A site has no left neighbour, so there is no extra boundary condition.

Eq. (17) is thus an exact eigenstate at $E = 0$. It is normalisable iff $\sum_n |v/w|^{2n} < \infty$, i.e. iff $|v| < |w|$, which is exactly the $\nu = 1$ phase of Eq. (10). Defining $|\psi_A(n)| = e^{-n/\xi}|\psi_A(0)|$ gives the localisation length

$$\xi = \frac{1}{\ln|w/v|}. \tag{18}$$

It diverges at the gap closing, as it must. The right end of a long chain supports the mirror-image state on B only. The zero mode is therefore pinned to a single sublattice and to $E = 0$ by chiral symmetry alone, and the amplitude ratio of neighbouring cells is exactly $-v/w$.

2.7 Exact spectrum of the open chain

The finite chain can be solved exactly, which gives a second derivation of both the edge state and its splitting. Label the cells $n = 1, \ldots, N$. The eigenvalue equations are

$$E\,a_n = v\,b_n + w\,b_{n-1}, \qquad E\,b_n = v\,a_n + w\,a_{n+1}, \tag{19}$$

with a_n, b_n the amplitudes on A_n, B_n. The open ends are imposed by extending the chain with two fictitious sites and requiring

$$b_0 = 0, \qquad a_{N+1} = 0. \tag{20}$$

Standing waves. Bulk solutions are $a_n = c_1 e^{ikn} + c_2 e^{-ikn}$, and the second equation then gives

$$b_n = \frac{1}{E}\big[h(k)\, c_1 e^{ikn} + h(-k)\, c_2 e^{-ikn}\big], \qquad E^2 = h(k)\, h(-k). \tag{21}$$

For real k, $h(-k) = h(k)^*$. Write $h = |h|e^{i\phi}$. The left boundary, $b_0 = 0$, gives $c_2 = -e^{2i\phi} c_1$. The right boundary, $a_{N+1} = 0$, then requires $\sin\big(k(N+1) - \phi\big) = 0$. Multiplying by $|h|$ and using $|h|\cos\phi = v + w\cos k$ and $|h|\sin\phi = w\sin k$,

$$v \sin\big(k(N+1)\big) + w \sin(kN) = 0. \tag{22}$$

This replaces the periodic condition $k = 2\pi m/N$. Each root $k \in (0, \pi)$ gives a pair of eigenvalues $\pm|h(k)|$.

Counting roots. Eq. (22) has N real roots in $(0, \pi)$ when $v(N+1) > wN$. Near $k = \pi$, with $k = \pi - q$, the left-hand side is $(-1)^N q\,[v(N+1) - wN] + O(q^3)$, so when $v(N+1) < wN$ its sign near $k = \pi$ changes and one real root is lost. The missing root moves off the real axis to $k = \pi + i\kappa$. Substituting into Eq. (22),

$$v \sinh\big((N+1)\kappa\big) = w \sinh(N\kappa). \tag{23}$$

At small κ the ratio of the two sides is $v(N+1)/(wN)$, and at large κ it grows as $(v/w)e^{\kappa}$, so a solution $\kappa > 0$ exists exactly when $v/w < N/(N+1)$. For a long chain this is the topological phase $v < w$, and the complex wavenumber is the edge state: $|e^{ikn}| = e^{-\kappa n}$.

Localisation length and splitting, again. For large N, Eq. (23) gives $e^{\kappa} \to w/v$, i.e. $\kappa = 1/\xi$ with ξ from Eq. (18). Keeping the first correction, with $r = v/w$,

$$e^{-\kappa} = r\big[1 + r^{2N}(1 - r^2) + \dots\big]. \tag{24}$$

For complex k the energy is $E^2 = h(k)h(-k) = (v - we^{-\kappa})(v - we^{\kappa})$. The first factor is small, $-v\, r^{2N}(1 - r^2)$, and the second is $(v^2 - w^2)/v$ to leading order, so

$$E^2 = r^{2N}(1 - r^2)(w^2 - v^2) \quad \Longrightarrow \quad |E| = w\,(1 - r^2)\, r^N, \tag{25}$$

which is Eq. (30) below, obtained here without assuming the form of the end states.

2.8 Finite-size splitting

On a chain of N cells the left state $|L\rangle$ (on A) and the right state $|R\rangle$ (on B) are no longer exact eigenstates. With $r = v/w < 1$ the normalised states are

$$\langle A_n|L\rangle = \sqrt{1 - r^2}\,(-r)^n, \qquad \langle B_{N-1-m}|R\rangle = \sqrt{1 - r^2}\,(-r)^m, \tag{26}$$

for $n, m = 0, \dots, N-1$, up to corrections of order r^{2N} in the normalisation.

Residual of $|L\rangle$. Every B row of $H|L\rangle$ vanishes by Eq. (17), except the last one, B_{N-1}, which would need the missing amplitude $\psi_A(N)$. There

$$H|L\rangle = v\, \psi_A(N-1)\, |B_{N-1}\rangle. \tag{27}$$

Effective 2×2 problem. In the basis $\{|L\rangle, |R\rangle\}$, chiral symmetry makes the diagonal elements vanish, and the off-diagonal element is

$$\langle R|H|L\rangle = v\, \psi_A(N-1)\, \langle R|B_{N-1}\rangle \tag{28}$$

$$= v\,(1 - r^2)\,(-r)^{N-1}. \tag{29}$$

The two states hybridise into $(|L\rangle \pm |R\rangle)/\sqrt{2}$ with energies $\pm\varepsilon$, where

$$\varepsilon(N) \simeq v\,(1 - r^2)\, r^{N-1} = w\Big(1 - \frac{v^2}{w^2}\Big)\Big(\frac{v}{w}\Big)^N. \tag{30}$$

The slope of $\ln\varepsilon$ against N is $\ln(v/w) = -1/\xi$.

Dimensional check. Under $H \to \lambda H$, i.e. $(v, w) \to (\lambda v, \lambda w)$, every eigenvalue scales by λ while v/w is unchanged. The prefactor of $(v/w)^N$ must therefore carry one power of energy. The often-quoted form $(1 - (v/w)^2)(v/w)^N$ is dimensionless and is only correct at $w = 1$.

2.9 Counting zero modes: the chiral index

Restrict H to a long chain and consider the zero-energy states localised near the left end. Since Γ commutes with H in the zero-energy subspace, these states can be split into N_A states on sublattice A and N_B on sublattice B. A chiral-symmetric perturbation can move a zero mode away from $E = 0$ only together with a partner of opposite chirality, because the eigenvalues must stay in $\pm E$ pairs and a state at $E \neq 0$ has equal weight on both sublattices. The difference

$$N_A - N_B \tag{31}$$

is therefore unchanged by any chiral-symmetric perturbation that keeps the bulk gap open. Bulk–boundary correspondence identifies it with the winding number, $N_A - N_B = \nu$ at the left end and $-\nu$ at the right end. For the SSH chain, Eq. (17) is the single A-state for $\nu = 1$.

2.10 Real-space winding number

With disorder the hoppings become

$$v_n = v + W_v\, r_n, \qquad w_n = w + W_w\, r'_n, \tag{32}$$

with r_n, r'_n independent and uniform on $[-\frac{1}{2}, \frac{1}{2}]$. Disorder in the hoppings preserves $\Gamma H \Gamma = -H$; disorder in on-site energies would not. Momentum is no longer a good quantum number, so ν must be defined in real space. We use the formula of Mondragon-Shem *et al.* [4],

$$\nu = \frac{1}{2L'} \operatorname{Tr}' \left[\Gamma Q [Q, X]\right], \qquad Q = \operatorname{sign}(H), \tag{33}$$

where X is the cell-position operator and Tr' runs over the central L' cells, away from the ends.

Flat-band operator. $Q = P_+ - P_-$, where $P_\pm$ project onto positive and negative energies. It has the same eigenvectors as H, eigenvalues ± 1, and $Q^2 = 1$. Because $\Gamma Q \Gamma = -Q$, it has the chiral block form

$$Q = \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix}, \qquad \Gamma = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{34}$$

and $Q^2 = 1$ means q is unitary.

Normalisation. Multiplying out,

$$Q[Q, X] = Q^2 X - QXQ = X - QXQ, \tag{35}$$

$$\Gamma Q[Q, X] = \operatorname{diag}\left(X - qXq^\dagger,\ -(X - q^\dagger X q)\right). \tag{36}$$

For a clean chain, q acts as a translation by ν cells, so $qXq^\dagger = X - \nu$ and $q^\dagger X q = X + \nu$ in the bulk. The two blocks then contribute $+\nu L'$ and $+\nu L'$, giving

$$\operatorname{Tr}' \left[\Gamma Q [Q, X]\right] = 2\nu L', \tag{37}$$

which fixes the factor $1/2L'$ in Eq. (33). With disorder Eq. (33) is no longer an integer for a single realisation, but its disorder average is quantised wherever the $E = 0$ states are localised.

2.11 Disorder-driven transition

At $E = 0$ the B rows of $H\psi = 0$ still decouple from the A rows:

$$v_n\,\psi_A(n) + w_n\,\psi_A(n+1) = 0 \quad \Longrightarrow \quad \psi_A(N) = \prod_{n=0}^{N-1} \Big(-\frac{v_n}{w_n} \Big) \psi_A(0). \tag{38}$$

Taking the logarithm turns the product into a sum of independent random variables,

$$\frac{1}{N} \ln \left| \frac{\psi_A(N)}{\psi_A(0)} \right| = \frac{1}{N} \sum_n \big(\ln|v_n| - \ln|w_n| \big) \xrightarrow{N\to\infty} \langle \ln|v| \rangle - \langle \ln|w| \rangle. \tag{39}$$

By the law of large numbers this is the (negative) Lyapunov exponent. The left end state decays, and the chain is topological, iff

$$\langle \ln|w_n| \rangle > \langle \ln|v_n| \rangle. \tag{40}$$

The transition is where the two log-averages are equal, and there the $E = 0$ localisation length $1/|\langle \ln|w| \rangle - \langle \ln|v| \rangle|$ diverges. In the clean limit this reduces to $|v| < |w|$.

Closed form for box disorder. For r uniform on $[-\frac{1}{2}, \frac{1}{2}]$, substitute $x = a + Wr$:

$$\langle \ln|a + Wr| \rangle = \frac{1}{W} \int_{a-W/2}^{a+W/2} \ln|x|\, dx. \tag{41}$$

With the antiderivative $F(x) = x \ln|x| - x$ (valid across $x = 0$, where the singularity is integrable),

$$\langle \ln|a + Wr| \rangle = \frac{F(a + W/2) - F(a - W/2)}{W}. \tag{42}$$

Solving $\langle \ln|v_n| \rangle = \langle \ln|w_n| \rangle$ with Eq. (42) gives the critical disorder without any fit.

Topological Anderson insulator. The log-average is not monotonic in W: for $W > 2a$ the interval contains $x = 0$ and $\langle \ln|a + Wr| \rangle$ is pulled down. If $v > w$ (clean chain trivial) and only v_n is strongly disordered, $\langle \ln|v_n| \rangle$ can drop below $\langle \ln|w_n| \rangle$ in a window of W. Disorder then *creates* a topological phase.

Equal disorder. With $W_v = W_w = W$ and $v < w$, the integrand $\ln|x|$ is averaged over two intervals of equal width centred at v and w. One can show that $\langle \ln|v + Wr| \rangle < \langle \ln|w + Wr| \rangle$ for every W, with the difference vanishing only as $W \to \infty$. The chain stays topological at all disorder strengths, but its $E = 0$ localisation length grows without bound.

3 Numerical methods

Each quantity is computed by a method that is exact in some limit, so that its error is either zero or a known discretisation error. No eigenvector is ever differentiated: the eigensolver returns each eigenvector with an arbitrary phase, so a finite difference $\partial_k u$ is meaningless.

3.1 Winding number by phase increments

On a closed grid $k_j = 2\pi j / N_k$,

$$\nu = \frac{1}{2\pi} \sum_{j=0}^{N_k - 1} \arg \frac{h(k_{j+1})}{h(k_j)}, \qquad k_{N_k} \equiv k_0. \tag{43}$$

Each increment is the principal value in $(-\pi, \pi]$. The sum of increments around a closed loop is exactly 2π times an integer, and that integer equals the continuum winding as long as no single step changes $\arg h$ by more than π. For $h = v + we^{ik}$ the steepest change of $\arg h$ is $w/|w-v|$ per unit k, so the condition is $N_k > 2|w|/|w-v|$, which fails only next to the gap closing. Exactly at $|v| = |w|$, $h(\pi) = 0$ and the routine returns `nan`.

3.2 Zak phase by Wilson loop

The continuum line integral of $A(k)$ is replaced by the phase of a product of overlaps,

$$\gamma = -\operatorname{Im} \ln \prod_{j=0}^{N_k - 1} \langle u_j | u_{j+1} \rangle, \qquad u_{N_k} \equiv u_0, \qquad N_k = 400. \tag{44}$$

To see that this is the right object, expand $\langle u_j | u_{j+1} \rangle = 1 + \delta k \, \langle u | \partial_k u \rangle + O(\delta k^2) = \exp(-iA\,\delta k) + O(\delta k^2)$; the product of exponentials gives $\exp\bigl(-i \oint A \, dk\bigr)$. Each u_j appears once as a bra and once as a ket, so a gauge change $u_j \to e^{i\theta_j} u_j$ multiplies the product by $\prod_j e^{-i\theta_j} e^{i\theta_{j+1}} = 1$. The discrete formula is therefore gauge invariant at every N_k, not just in the continuum limit. For the SSH chain $\gamma = \pm\pi$ or 0 is fixed by symmetry, so the discrete value is exact as well.

3.3 Spectrum of the open chain

The open chain is a real symmetric tridiagonal matrix, diagonalised with the LAPACK symmetric eigensolver (Householder reduction followed by divide-and-conquer). Its eigenvalues have absolute error $O(\epsilon_{\text{mach}} \|H\|)$, which is why the zero modes appear at $\sim 10^{-16}$.

3.4 Edge states by chiral projection

In the topological phase the two eigenvectors closest to $E = 0$ are the hybridised pair $(|L\rangle \pm |R\rangle)/\sqrt{2}$ at $\pm\varepsilon$, with weight on both ends. To separate them, let V be the $2N \times 2$ matrix of the two eigenvectors and form

$$\Gamma_{\text{eff}} = V^\dagger \Gamma V. \tag{45}$$

Diagonalising this 2×2 matrix gives eigenvalues close to ± 1; the $+1$ eigenvector, mapped back with V, is the state on A (left end), and the -1 eigenvector is the state on B (right end). States are identified by sublattice weight rather than by eigenvalue index, which would be arbitrary for a nearly degenerate pair.

3.5 Localisation length and splitting fits

ξ is obtained by a linear least-squares fit of $\ln |\psi_A(n)|$ against n on a 100-cell chain, using cells where the amplitude is above round-off. The splitting slope is a linear fit of $\ln \min |E|$ against N; the prefactor is the ratio $\varepsilon/(v/w)^N$ at $N = 30$, where corrections of order $(v/w)^{2N}$ are negligible.

3.6 Real-space winding and disorder averaging

For each disorder realisation, H is diagonalised as $H = U \operatorname{diag}(E) U^\dagger$ and the flat-band operator is built as

$$Q = U \operatorname{diag}\bigl(\operatorname{sign} E\bigr) U^\dagger. \tag{46}$$

Eq. (33) is then evaluated with the trace restricted to the central $L' = N/2$ cells, so that the ends, where the edge states live, do not contribute. Each point averages 100 independent realisations; error bars are standard errors of the mean, $\sigma/\sqrt{100}$. The realisations are split over processes with an independently seeded random stream per chunk, so the result does not depend on the number of workers.

3.7 Critical disorder

The transition is located without simulation by evaluating Eq. (42) for both hoppings on a grid of 20 001 values of W up to $W = 10$, finding the sign changes of $\langle \ln |v_n| \rangle - \langle \ln |w_n| \rangle$, and refining each by linear interpolation between neighbouring grid points. A sign change from positive to negative marks entry into the topological phase and the reverse marks exit, so the topological Anderson window appears as two crossings. The antiderivative is evaluated with the convention $F(0) = 0$, which is the limit $x \ln |x| \to 0$.

3.8 Finite-size scaling

For the scaling study the real-space winding is computed for $N = 50, 100, 200, 400, 800$ cells, always tracing over the central half. Inside a phase the disorder average approaches the quantised value as N grows; near the transition it moves away from the quantised value. A crossing of the curves for different N therefore brackets the transition, independently of Eq. (42). The cost of each realisation is dominated by the full diagonalisation needed for Q, which scales as $O(N^3)$.

4 Verification

The checks fall into five categories. Each compares a numerical result with a statement derived in Section 2, so a failure points directly at either the theory or the method.

1. Exact identities. Several results hold exactly for any parameters and so must be met to round-off:

- The numerical bands agree with $\pm|h(k)|$, and the minimum gap equals $2|v - w|$.
- Both the Bloch and the open-chain Hamiltonians anticommute with Γ.
- The flat-band operator satisfies $Q^2 = 1$.
- On the open chain the edge-state amplitudes of neighbouring cells are in the ratio $-v/w$ exactly, as Eq. (17) requires. This is a sharper test than the fitted ξ, because it is a pointwise statement.

2. Quantisation. The winding number must be an integer to round-off away from $v = w$, take the value predicted by Eq. (10) on each side, and be undefined at $v = w$. The Zak phase must equal $\pi\nu$ modulo 2π.

3. Invariance. The Wilson loop is evaluated a second time after multiplying every u_j by an independent random phase. Since the discrete formula is gauge invariant by construction, the two results must agree to round-off; a method that differentiated eigenvectors would fail this test.

4. Asymptotic laws. The edge state, splitting and localisation length obey asymptotic forms, so they are tested in the regime where the neglected terms are small:

- zero modes at machine zero for $v < w$, none below the band edge for $v > w$;
- the zero modes are sublattice-pure after chiral projection;
- the fitted ξ against Eq. (18);
- the slope of $\ln \varepsilon$ against $\ln(v/w)$, and the prefactor against $w(1 - v^2/w^2)$. The test includes $(v, w) = (1, 2)$, where the two candidate prefactors differ by a factor of two, so it discriminates between them rather than merely being consistent with one.

5. Cross-checks between independent methods.

- On a clean chain the real-space winding of Eq. (33) must reproduce the Bloch winding. The two are computed from unrelated objects (a matrix sign function in real space and a phase sum in k-space).

- The closed form Eq. (42) is compared with a Monte-Carlo average of $\ln|a+Wr|$ over many samples, within the statistical error.
- With weak disorder the averaged winding must remain 1; with very strong disorder it must vanish; with equal disorder on both hoppings the two log-averages must never cross.

5 Results

5.1 Bulk bands and invariants

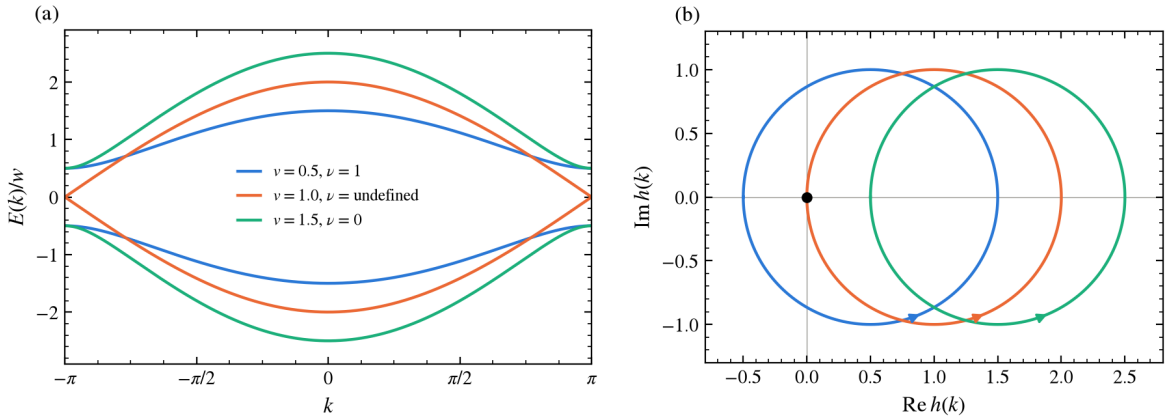


Figure 1: (a) Bulk bands for $w = 1$ and $v = 0.5$ (topological), 1.0 (critical), 1.5 (trivial); the gap closes at $k = \pi$ when $v = w$. (b) $h(k)$ in the complex plane: the loop encloses the origin only for $v < w$, giving $\nu = 1$.

Figure 1 shows the two ingredients of the bulk picture: the gap $2|v - w|$ at $k = \pi$ and the circle traced by $h(k)$. Figure 2 shows that both invariants are exactly quantised and change only at $v = w$.

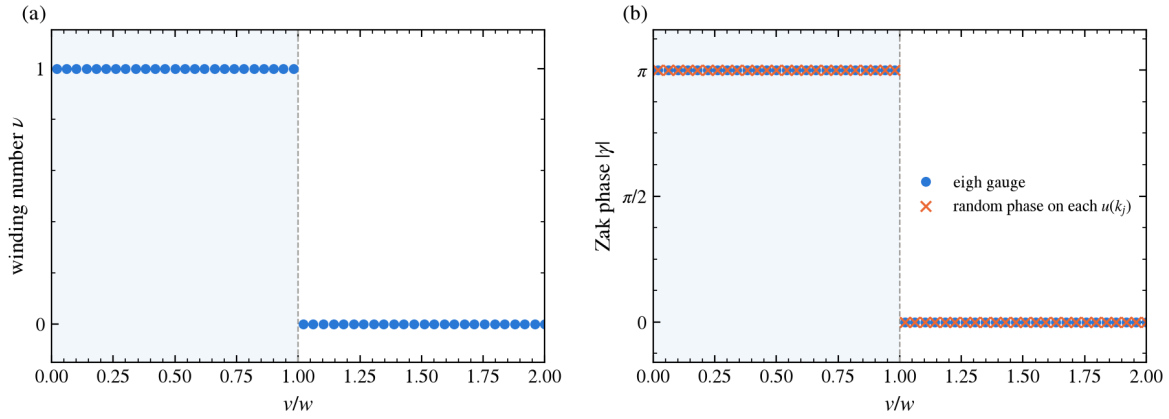


Figure 2: (a) Winding number versus v/w, an integer to 2.5×10^{-16} that jumps only at the gap closing. (b) Zak phase from a 400-point Wilson loop: $|\gamma| = \pi$ for $v < w$, 0 for $v > w$ (error $\leq 1.3 \times 10^{-15}$). Crosses: same calculation with a random phase on every $u(k_j)$, identical to 2.5×10^{-15}.

5.2 Open chain and edge states

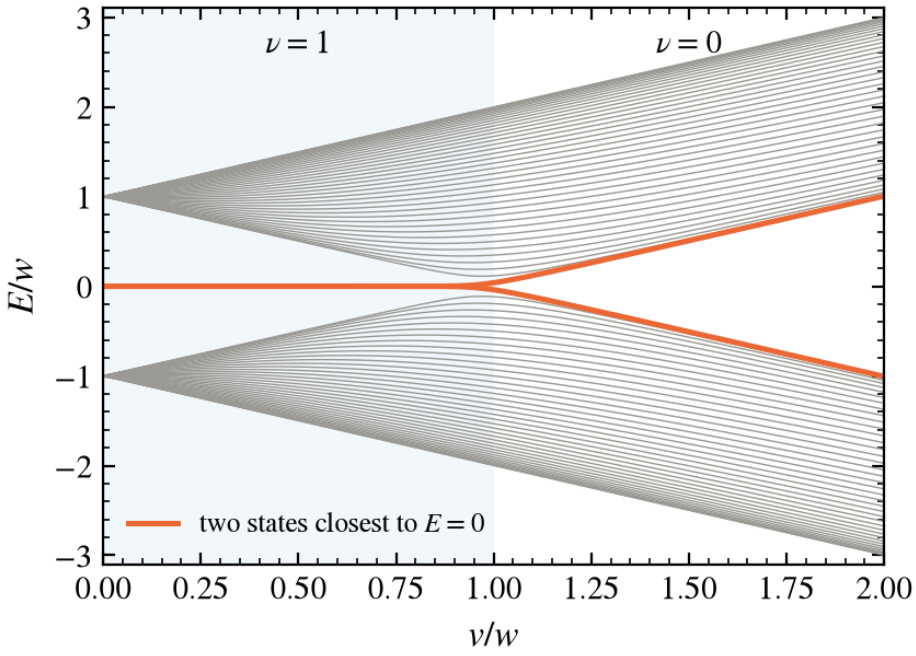


Figure 3: Spectrum of a 40-cell open chain versus v/w. In the shaded topological region two states are pinned at $E = 0$ (2.2×10^{-16} at $N = 60$, $v/w = 0.5$). For $v > w$ the lowest states sit at the band edge ($\min |E| = 0.504 \approx |v - w|$ at $v/w = 1.5$).

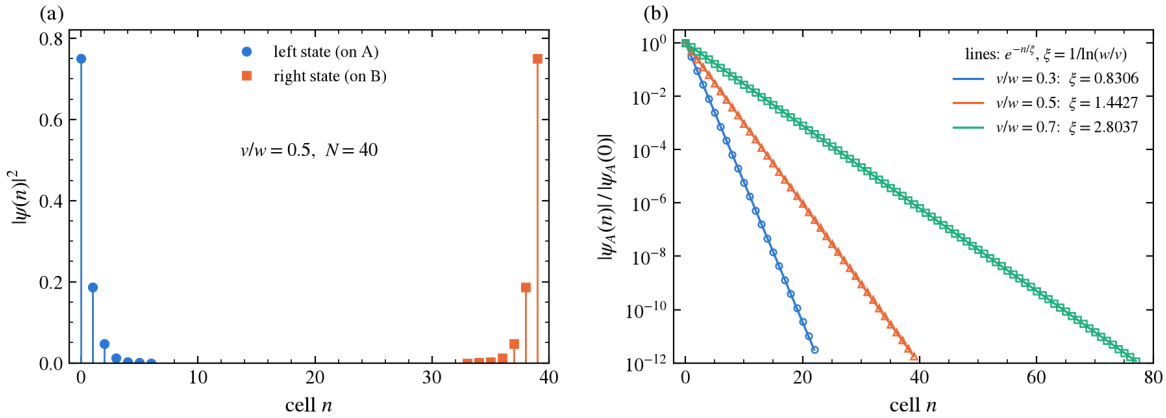


Figure 4: (a) The two zero modes after chiral projection: left state on A only, right state on B only. (b) $|\psi_A(n)|$ on a 100-cell chain (symbols) against $e^{-n/\xi}$ with $\xi = 1/\ln(w/v)$ (lines).

Table 2: Fitted versus analytic localisation length. Relative errors are round-off and vary with the BLAS build, so only the bound is quoted.

v/w	ξ fitted	$1/\ln(w/v)$	relative error
0.3	0.830584	0.830584	$< 2 \times 10^{-13}$
0.5	1.442695	1.442695	$< 2 \times 10^{-13}$
0.7	2.803673	2.803673	$< 2 \times 10^{-13}$

The chirally projected state on a chain much longer than ξ is the analytic zero mode: neighbouring amplitudes are in the exact ratio $-v/w$, so the agreement is at machine precision rather than merely within a fitting tolerance.

5.3 Edge-state splitting

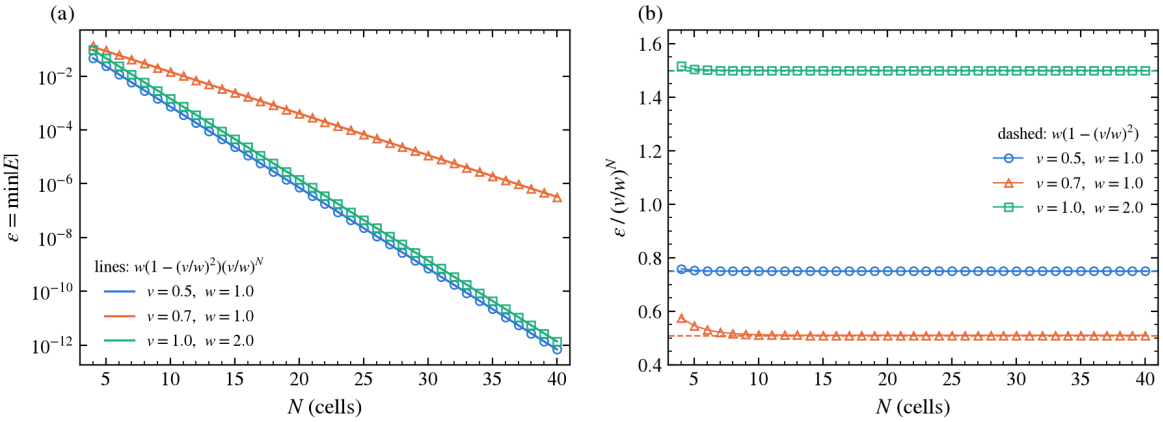


Figure 5: (a) $\varepsilon = \min|E|$ against chain length: symbols from exact diagonalisation, lines from Eq. (30). (b) $\varepsilon/(v/w)^N$ converges to the dashed prefactor within about 10 cells.

Table 3: Splitting slope and prefactor. The $(1, 2)$ row distinguishes the two candidate prefactors.

(v, w)	fitted slope	$\ln(v/w)$	$\varepsilon/(v/w)^N$, $N = 30$	$1-(v/w)^2$	$w(1-(v/w)^2)$
$(0.5, 1)$	-0.69316	-0.69315	0.750	0.750	0.750
$(0.7, 1)$	-0.35706	-0.35667	0.510	0.510	0.510
$(1, 2)$	-0.69316	-0.69315	**1.500**	0.750	**1.500**

The slope reproduces $\ln(v/w)$ to four digits. The $(1, 2)$ row gives 1.500, as predicted by Eq. (30), and excludes the dimensionless prefactor 0.750.

5.4 Real-space winding under disorder

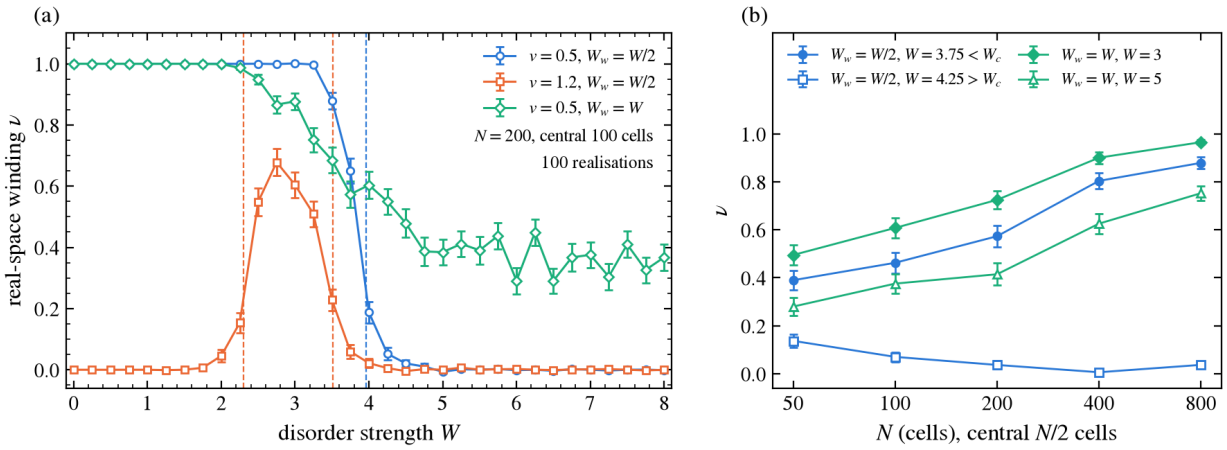


Figure 6: (a) Disorder-averaged real-space winding on a 200-cell chain (trace over the central 100 cells), 100 realisations per point. Dashed lines: predicted transitions from Eq. (42), in the colour of the matching curve. (b) Finite-size dependence for $N = 50$ to 800.

Topological chain, $v = 0.5$, $W_w = W_v/2$. $\nu = 1.000 \pm 0.003$ up to $W = 3.25$, even though the hoppings already change sign for $W > 1$. It then falls through 0.88 ($W = 3.5$), 0.65 (3.75) and 0.19 (4.0) and vanishes by $W \approx 4.5$; the prediction is $W_c = 3.959$. Finite-size scaling (Fig. 6b) sharpens the step: at $W = 3.75$, ν grows from 0.39 to 0.88 as N goes from 50 to 800, while at $W = 4.25$ it falls from 0.14 to 0.04.

Topological Anderson insulator, $v = 1.2$. The clean chain is trivial. Disorder alone drives it topological: ν rises to 0.68 ± 0.04 at $W = 2.75$, inside the predicted window $2.299 < W < 3.515$, and returns to zero at stronger disorder. At $N = 200$ the plateau does not reach 1 because the window is narrow and both its edges are critical.

Equal disorder, $W_v = W_w = W$. For $v = 0.5$, $w = 1$ the log-averages never cross, so the chain is topological at every disorder strength. The drop of the green curve in Fig. 6a is a finite-size effect: ν climbs from 0.50 to 0.97 at $W = 3$ and from 0.28 to 0.75 at $W = 5$ as N grows from 50 to 800.

A single condition, $\langle \ln |v_n| \rangle = \langle \ln |w_n| \rangle$, therefore accounts for the clean-to-trivial transition, the re-entrant topological window, and the absence of a transition for equal disorder.

6 Discussion

Why the winding cannot change without closing the gap. ν counts encirclements of the origin by $h(k)$; a continuous deformation can change it only by passing $h(k)$ through the origin, i.e. making $E(k) = 0$. With disorder the role of the gap is played by the mobility gap: ν changes only where the $E = 0$ localisation length diverges, which is exactly the Lyapunov condition derived above.

Breaking chiral symmetry. A staggered potential $\pm m$ adds $m\sigma_z$ to $H(k)$, which commutes with Γ. The spectrum loses its $\pm E$ symmetry, the edge states move to $E = \pm m$, $\mathbf{d}(k)$ leaves the plane, and ν is no longer quantised.

Unit-cell dependence of the Zak phase. Choosing the cell (B_n, A_{n+1}) swaps $v \leftrightarrow w$ and exchanges $\gamma = \pi$ and 0; moving the origin inside the cell shifts γ continuously. The Zak phase is a polarisation, defined only once an origin is fixed. Physical are differences in γ, and the edge-state count for a given termination [5].

What "protected" means. For perturbations that keep chiral symmetry and keep the $E = 0$ states localised, the zero modes stay at exactly $E = 0$, sublattice-polarised and exponentially localised. In Fig. 6 this holds up to $W \approx 3.25$. Protection fails when chiral symmetry is broken, when ξ diverges ($W \to W_c$), or when the chain is short enough that the ends hybridise.

Limitations. (i) Disorder averages use 100 realisations; the topological Anderson plateau would need $N \gg 200$ to reach $\nu = 1$. (ii) The Lyapunov criterion is a thermodynamic-limit statement; at finite N the crossover is broadened by $\xi(E = 0) \sim 1/|\langle \ln |w| \rangle - \langle \ln |v| \rangle|$. (iii) Only uniform box disorder is studied; Eq. (42) would need replacing for other distributions.

7 Conclusion

Every analytic statement about the SSH chain (integer winding, $\gamma = \pi\nu$, sublattice-pure zero modes with $\xi = 1/\ln|w/v|$, exponential edge splitting) has been reproduced to machine precision by gauge-invariant numerics. The edge-splitting prefactor follows from the overlap of the two end states and is $w(1 - v^2/w^2)$. With disorder, the zero-energy transfer matrix gives a closed-form criterion that predicts the topological transition and the topological Anderson insulator, and finite-size scaling shows that equal disorder on both hoppings never trivialises the chain. The same toolkit (Wilson loops, chirally projected edge states, real-space invariants) underpins the Berry-phase, non-Hermitian SSH and non-Bloch projects in this repository.

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